

PART

Three

Inflation-Linked Debt

Inflation-Indexed Bonds

23.1 INTRODUCTION

The importance of inflation-linked debt (instead of, say, debt linked to the weather¹) lies in the basic meaning of the word ‘fixed income security’. While a nominal bond provides a fixed income in the sense of a fixed cashflow, what investors are usually interested in is not income *per se*, but consumption. The advantage of a fixed income is that if prices are stable (or at least move predictably), it will provide a steady (or at least known) stream of goods. For most of human history, this happy state of affairs has indeed been present and therefore fixed income securities have established themselves in the economy. However, at times prices can move quite sharply and unpredictably and thereby call into question the usefulness of a fixed nominal income. At such times, borrowers find more willing investors if they issue debt where cashflows are explicitly linked to prices. In doing so, they commit to compensate the lender for any loss in purchasing power experienced by the currency in which the bond cashflows are being paid. The most ready explanation for the existence of inflation-linked debt therefore is uncertainty about price developments.

The first inflation-linked bond known to the author was issued in what is now the US State of Massachusetts in 1780. At the time, the guardian of the currency used in Massachusetts was the Bank of England, which may be presumed to have been somewhat unconcerned with the purchasing power of bond coupons paid in what was a far away part of Britain that was waging its War of Independence. The bond in question, therefore, while nominally paying in pounds sterling, actually paid in what one would now call a basket of consumer goods. The statement of currency is therefore almost redundant and indeed kept open through the use of ‘to be paid in the then current Money of said State’. Modern inflation-linked bonds do no longer spell out the underlying basket and instead refer to well-known indices published by respected statistics offices. However, this bond actually has a structure that is virtually identical to what is now generally known as the TIPS structure, so in order to prepare for working with the modern security, it’s useful to spend some time analysing this one.

¹Debt linked to weather exists but is a niche product.

The inscription of the bond, for one sample, reads:

In behalf of the State of Massachusetts-Bay, I the Subscriber do hereby promise and oblige Myself and Successors in the Office of TREASURER of said STATE to pay unto [name] or to his order, the SUM of Four hundred & seventy Pounds 8/9 on or before the First Day of March, in the Year of our Lord One Thousand Seven Hundred and eighty one with Interest at *Six per Cent.* per Annum: Both Principal and Interest to be paid in the then current Money of said State, in greater or less Sum, according as Five Bushels of CORN, sixty-eight Pounds and four-seventh Parts of a Pound of BEEF, Ten Pounds of SHEEPS WOOL, and Sixteen Pounds of SOLE LEATHER shall then cost, more or less than *One Hundred and Thirty Pounds* current Money, at the then current Prices of said ARTICLES — This SUM being THIRTY-TWO TIMES AND AN HALF what the same Quantities of the same Articles would cost at the Prices affixed to them in a Law of this STATE made in the Year of our Lord One Thousand Seven Hundred and Seventy-seven, intituled, '*An Act to prevent Monopoly and Oppression*': The current Prices of said Articles, and the consequent Value of every Pound of the SUM herein promised, to be determined agreeable to a LAW of this STATE, intituled, '*An ACT to provide for the Security and Payment of the Balances that may appear to be due by Virtue of a Resolution of the GENERAL ASSEMBLY of the Sixth of February One Thousand Seven Hundred and Seventy-nine, of this STATE'S Quota of the CONTINENTAL ARMY, agreeable to the Recommendation of CONGRESS, and for Supplying the TREASURER with a SUM of Money for that Purpose*'.

While this may sound somewhat obscure at first, it is relatively simple. First, the obligation is for the Treasurer of the State of Massachusetts to pay a certain sum to the person inscribed on the bond, or any other person if the original owner so wishes. The latter clause makes the obligation transferrable and thereby a bond. Without the addition of 'or to his order', the bond holder would have to claim the payment in person, which, given that this is soldiers' scrip, would devalue the obligation significantly. The payment is for 470 8/9 pounds (at the time still English pounds) plus 6% interest per annum, both adjusted for a price level. Assuming for simplicity that the bond pays interest for exactly one year, the final obligation will be 499.14 pounds, adjusted for the price level.

The price level adjustment to be made on payment is documented as follows. First, the prices of a bushel of corn (p_c), a pound of each of beef (p_b), sheeps wool (p_w), and sole leather (p_l) are to be established according to a law passed in 1779 which deals with financing the contribution of Massachusetts to the continental army for fighting the British. Second, the quantity:

$$I_M = 5p_c + 68\frac{4}{7}p_b + 10p_w + 16p_l \quad (23.1)$$

is to be calculated. The payment to be made is then 'a greater or less sum . . . than 130 pounds current money'. In other words, the payment obligation will move up and down

together with I_M and if $I_M = I_0 = 130$ pounds, then the sum to be paid will be the one calculated above, i.e., 499.14 pounds. Written as a formula, the treasurer would have to pay the sum of S pounds where:

$$S = \frac{I_M}{I_0} 499.14 \quad (23.2)$$

The ratio I_M/I_0 is what is now referred to as an index ratio, i.e., the ratio of current to historical price level. Note that the expression ‘in the then current money of said state’ is almost superfluous since the prices, and hence I_M , would presumably be measured in current money. Using hindsight and calling the ‘then money’ dollar,

$$S[\text{dollar}] = \frac{I_M[\text{dollar}]}{130[\text{pounds}]} 499.14[\text{pounds}] \quad (23.3)$$

so that the pound can be eliminated from the equation and only the dollar remains.

The convention to adjust payments by multiplication with an index ratio is the TIPS convention referred to above.

As a last send-off to this interesting document, we note that Massachusetts apparently passed a law fixing the prices of these items in 1777, but within the span of 3 years inflation had eroded the currency so much that prices had increased 32.5 times, i.e., annual inflation was running at 220 percent.

Seeing inflation-linked debt only as the last resort of debtors in countries with high and/or unstable inflation is misleading, however. An alternative is to issue bonds in a foreign, more stable currency. While indeed many emerging market economies issue inflation-linked debt to address the issue of currency stability worries, the biggest issuers of inflation-linked bonds are large economies with relatively stable rates of inflation. Therefore, there must be other reasons to issue and invest in such debt.

For sovereign issuers with progressive income tax systems, an important incentive to issue is the so-called tax creep. If all prices were doubled overnight and with it, all salaries were doubled, too, so that overnight the currency has been devalued by 50% in real terms, assuming for argument's sake with full compensation for savers. If all taxes were proportional to prices and wages, all these things taken together should not make a difference to the economy. With progressive income tax, however, the price doubling means that everybody would be in a higher tax band and as a result, the government's tax take as a share of gross domestic product would increase. Of course, the government would eventually be forced to adjust the tax bands upwards to take into account the new price level, but this is generally done retroactively. Overall, therefore, governments with progressive income tax have a long position in inflation and, depending on tax progression, this can be significant. Of course, tax officials are aware of this fact and budget projections incorporate tax creep. However, because the state treasuries and the central banks of the larger economies tend to be more or less independent, inflation is not under the control of the government and the government may therefore be inclined to hedge the long inflation position brought about by tax creep through a short position created by selling inflation-index bonds.

An increasingly important second reason is that sometimes there is strong demand from investors for inflation-linked debt, leading to a valuation effect known as the inflation risk premium. A number of investors have explicitly inflation-linked liabilities, such as insurance contracts or pension promises, so they are interested to hedge this exposure. More importantly, because the performance of nominal and inflation-linked debt is not perfectly correlated, inflation-linked debt can be seen as a different asset class which is interesting for diversification purposes.

23.1.1 Cashflows of inflation-linked bonds

There are 2 structures that are commonly used in the market to compensate investors for inflation. The apparently simpler of the 2, called the TIPS structure, pays fixed coupons, but adjusts every cashflow for inflation. This is done through multiplying each cashflow, be it a coupon or redemption flow, or the invoice price for settling a bond purchase, by a so-called index ratio R_t defined as the ratio of the reference index I_t (explained below) for the settlement date t and a fixed base index I_0 , i.e.,

$$R_t = \frac{I_t}{I_0} \quad (23.4)$$

There is no particular reason to prefer one I_0 over any other because only the change in I_t affects the purchasing power, but in most markets the convention is to use the reference index for the first settlement date t_F of the bond so that $R_F = 1$. An exception is Sweden where most bonds use the same I_0 . Because the adjustment for inflation means that all coupons have effectively a constant purchasing power, they are referred to as real coupons and the bonds in general as real bonds.

The reference index can technically not be the inflation index itself because measuring and publishing inflation numbers is costly and takes time. Daily levels are therefore not available. If one were to rely on inflation indices only, I_t would change only once a month. In most markets, therefore, interpolation of lagged inflation numbers is used. In this setup, the reference index is based on the theoretical inflation level present at a fixed time before the settlement date of the relevant cashflow. This period is the inflation lag. When no actual inflation level for this day has been published, the theoretical level is calculated by linearly interpolating between the two closest published values, as shown in Figure 23.1. The minimum length of the lag period is therefore the time it takes to publish a new index value, plus about one month.

In most markets, R_t is floored at the level of R_0 , at least for the purpose of calculating the redemption cashflow. Note that the bond from 1780 cited above does not have such a floor because it uses the phrase ‘in a greater or less sum’.

An important point to realise about the TIPS-style adjustment of cashflows is that it is exactly identical to the treatment of cashflows of a foreign currency investment. In this setting, I_t plays the role of an FX rate. The foreign currency here is not one that exists in the real world. Instead, it is a hypothetical currency defined as having constant purchasing power in terms of the inflation index used for adjusting the cashflows. This analogy is extremely fruitful for understanding return and risk measures for index-linked bonds. It also explains the near-equivalence between foreign currency issuance and index-linked debt for small and open economies.

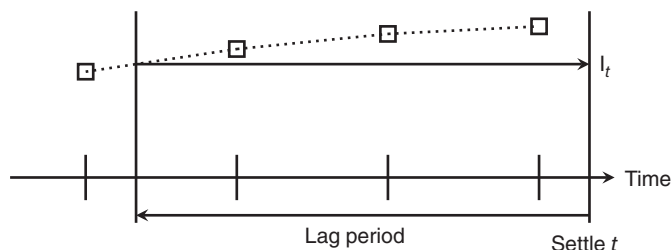


FIGURE 23.1 Reference index interpolation. The index level applied at time t is the interpolated level of the actual index at a given lag.

Because TIPS-style inflation-linked bonds adjust every cashflow for inflation, the loss of purchasing power on the principal is returned at the maturity of the bond. This means that the inflation-compensation component of interest rates is largely absent from the coupon payments and the coupons paid by TIPS-style inflation-linked bonds accordingly usually are lower than those paid by nominal bonds but the principal repayment is higher.

The alternative to the TIPS-style inflation link, used mostly in the corporate bond market and for the Italian retail BTP Italia, is to have a floating rate coupon defined as rate of inflation plus a spread. This type of adjustment, known as pay-as-you-go, means that coupons are usually higher than those of TIPS-style inflation-linked bonds, but the principal is repaid at par. Because the coupon rate is determined at the start of the coupon period, investors face some inflation risk but settlement is simpler than for TIPS-style bonds.

Given a term structure of inflation expectations, it is possible to compare the pricing of TIPS-style and floating rate note inflation-linked bonds, so at the outset there is no difference between the two types of link. However, the risk characteristics of the two types of bond are of course very different.

In the standard text [29], the TIPS convention is introduced as capital-indexed bonds while the floating rate note structure is called interest-indexed bond. This chapter deals almost exclusively with TIPS-style bonds because, despite their simple structure, they are more complex to analyse than pay-as-you-go schemes, and they are by far the most common.

Euro-area inflation-linked bonds tend to be linked not to the overall inflation index (usually euro-area HICP or, in the case of some French linkers, French CPI), but the index with tobacco removed from the index basket. The origin of this custom is the French loi Évin which bans tobacco advertising and promotion. France was the first euro-area country to issue inflation-linked bonds in large volumes². Later issuers found

²It is not entirely unfair to argue that retail tobacco prices are mostly taxes and it would be counter-intuitive for a government to pay out on government debt a cost that it imposes on its citizens. That being said, the French famously manipulated historical photographs to eliminate cigarettes from the images of well-known personalities at the same time as this convention was established. A health-related aspect of this convention can therefore not be ruled out.

that the inflation swap market had developed most liquidity in the ex-tobacco indices and followed this lead.

23.1.2 Quotation of index-linked bonds

Inflation-linked bonds are usually quoted with a clean price and net of inflation adjustment. In other words, the quoted price will, to first order, not change due to passage of time (affecting accrued interest) or inflation. To the extent that clean prices are a natural way of quoting bonds because they exclude predictable performance elements from the price quote, also excluding inflation, is a sensible convention. Despite, or perhaps because of, this the UK market quotes inflation-linked bonds including inflation.

The nominal prices of inflation-linked bonds are therefore given by scaling the real price by the index ratio applicable to the settlement date t , i.e.,

$$P(t) = R_t \hat{P}(t) \quad (23.5)$$

The invoice price, i.e., the consideration paid for an inflation-linked bond for settlement t is then:

$$V = NR_t(\hat{P} + Accr_t) \quad (23.6)$$

where accrued interest is calculated as it would be for a nominal bond.

In the FRN-style inflation-linked bonds, the invoice price is simply:

$$V = N(\hat{P} + Accr_t) \quad (23.7)$$

but the accrued interest is calculated from a floating-rate coupon that includes an inflation component.

23.2 REBALANCING, REBASING AND REVISION OF CPI INDICES

A CPI index is a chain-weighted average price of consumption items with the weights determined by their share in the consumption expenditure of individuals. As consumption patterns change over time, these weights also need to be revised. In addition, changes in fashions and technology also require adjustments in the items that are used to sample the prices of certain consumption items. For instance, flat-screen television sets used to be luxury items but have now completely replaced CRT-based devices, and mobile communication plays a much larger role than fixed-line telephony.

The statisticians compiling a CPI respond to such changes in two ways. The first are constant small adjustments in sampling methodology and sample items. Such adjustments tend to have only a marginal effect on the inflation rate. Because CPIs are chain-weighted, the replacement of one item with another has no impact on the level of the index, while the impact of the replacement on the rate of change of the CPI

depends on the item weight and the difference in the rate of price change between the old and new item. For regular adjustments, both of these numbers tend to be small, although on occasion they can be substantial.

Over longer horizons, it is necessary to revise weights in a more fundamental way by reassessing actual spending habits. These are sampled by consumer expenditure surveys which are costly to conduct. Such surveys therefore do not occur very often, roughly every 5 years or so, depending on the country. Consumer expenditure does not exactly match consumption expenditure because people spend cash on non-consumption items such as transfers (e.g. child support) or savings (including pensions and capital life insurance), and not every spending item might be included in the CPI index. For instance, a number of consumer price indices do not include the cost of owning a home (the so-called owner-equivalent rent) due to difficulties in accurately measuring this cost³. Certain cash spending items may, on the other hand, also relate to other consumption items. School fees, for instance, may include food items that are unrelated to education itself. Changing the measurement basket for the inflation index is known as rebalancing.

When index weights are revised on a broad basis, statistics offices may choose to let the index start from a new level. This is known as an index rebasing. In Japan, the quinquennial index revisions include a rebasing where the new index is scaled to that its average reading over the twelve months of the base year is equal to 100. Japanese CPI may therefore be referred to as, for instance, the ‘base-2010 CPI’ to distinguish it from the ‘base-2015 CPI’ with different index weights *and* a different base level. To the extent possible, a CPI index can be backfilled with weights of a later index vintage which makes indices comparable across time. Because the weight revision involved in a rebasing simultaneously affects the weights of many items, it can have a measurable impact on the inflation rate. Figure 23.2 shows the difference in the cumulative price change of the base-2010 and base-2015 nationwide all-items Japanese CPI between 2005 and 2015, the last year for which the base-2010 index was produced. Although quantisation noise disturbs the comparison, a different run rate is evident.

For inflation-linked bonds, index rebasing has two consequences. The first is that after a few years, the index vintage in which the base CPI is measured is discontinued. This means that the base index needs to be adjusted to the new index vintage. This could be done by either resetting the base index to the back-filled value of the new vintage index applicable to the date the original base index was set, or chaining the old and new index together during a transition period and converting the index ratio with the ratio of old and new index during that period. Specifically one uses:

$$I_{old}(t) = I_{new}(t) \frac{< I_{old} >}{< I_{new} >} \quad (23.8)$$

³Real estate transactions do not occur frequently or transparently, and part of the cost of purchasing real estate is related to the financing cost of mortgages, rather than the cost of property as such.

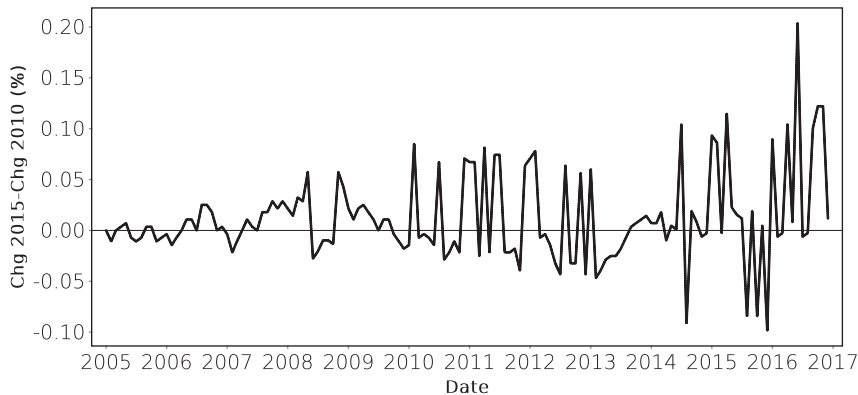


FIGURE 23.2 Difference between the Japanese CPI (all items) changes since January 2005 for the base-2010 and base-2015 indices. Note that index levels are reported with one decimal so quantisation noise is in the order of 0.1%. Source: MIC, e-stat.co.jp. Licenced under CC BY 4.0.

where the averages $\langle \cdot \rangle$ are taken over the base year of the new index vintage. In Japan this implies that $\langle I_{new} \rangle = 100$. When indices are rebased more than once, such adjustments can be chained. Note that the value of the fraction never changes.

This method has the advantage of leaving historical index ratios unchanged. Some data providers, like Bloomberg, incorporate this adjustment in the base CPI. This means that the base CPI of a given bond appears to change, contrary to how indexation has been introduced here. However, the value of the index ratio for any given day is unchanged by this approach.

The second consequence of rebasing is that the near-term run rate of inflation can be affected, which should then affect the fair value of breakeven inflation. This potential effect on breakeven is why index-linked bond analysts take a keen interest in rebasings which would otherwise be a rather niche subject⁴.

In other jurisdictions, such as Europe and the US, large-scale index weight changes are reflected not through the publication of a rebased new index vintage with a new base year, but through a revision of the existing index. In essence, past values of the CPI are recalculated using the new methodology, and scaled such that the current value of the index remains unchanged. While this approach avoids the potential confusion of multiple base-year indices, it breaks the idea that a past index level is immutable.

23.3 INFLATION SEASONALITY

Inflation figures are generally quoted in the press as annual rates of change and this is for most purposes the appropriate way of referencing price stability. Central banks also

⁴This statement should be taken in relative terms. Inflation-linked bonds are not exactly mainstream.

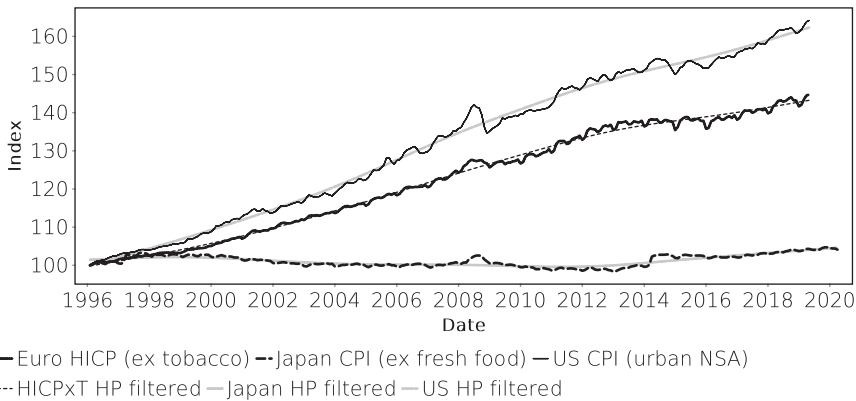


FIGURE 23.3 German, US and Japanese inflation indices over time. Sources: Data from ECS, BLS, MIC.

express price stability targets in terms of annual rates of change, with a 2% annual rate of change being a usual target point.

This convention obscures, however, persistent seasonal patterns in inflation. Figure 23.3 shows the main German, US, and Japanese price indices together with a Hodrick-Prescott (HP) filtered version that removes short-term fluctuations. The HP filter is used here for simplicity rather than accuracy⁵ and realistic seasonality adjustments usually use more complex filters such as the X12-ARIMA filter [76] which better reflect the impact of, for instance, movable feasts such as Easter.

The average differences between the actual monthly inflation readings and the smoothed index levels provides an estimate of the seasonality effects present in inflation⁶. Using the data in Figure 23.3, one finds the seasonal deviations (expressed as average percentage deviations from trend) shown in Figure 23.4.

Notably, the seasonal fluctuation in price levels is of an order of magnitude that is comparable to that of the underlying inflation itself. The Japanese CPI excluding fresh food in particular shows a seasonal variability well in excess of the long-term average of the inflation trend. This means that over any reasonable active investment horizon the inflation return of Japanese inflation-linked bonds is tied to the, more or less predictable, seasonal fluctuation in prices rather than the underlying inflation trend. Even for other inflation markets, the seasonal patterns are significant sources of returns.

The large seasonal fluctuations of inflation indices are driven by more or less predictable events such as seasonal sales campaigns and holiday dates. Figure 23.5 shows

⁵The main epistemological drawback of the HP filter is that adding new data can change the filtered value of older observations. This problem has not hurt its ubiquity in econometrics. For the purposes of the present analysis it is sufficiently accurate.

⁶There are two sources of errors in this estimate. The first is that over a finite observation period the effects of one-off events, such as strikes or transport disruptions, may not be averaged out completely. At the same time, longer observation periods may obscure time-variant phenomena such as the 'Black Friday' sales that are a recent phenomenon in retail pricing.

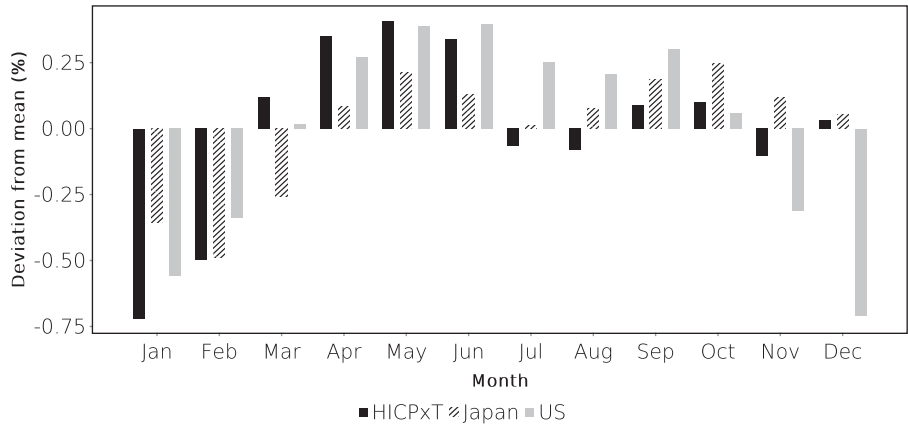


FIGURE 23.4 Seasonality factors (percentage deviations from trend developments) for the 3 inflation indices shown earlier. The trend is the HP-filtered monthly data shown earlier.

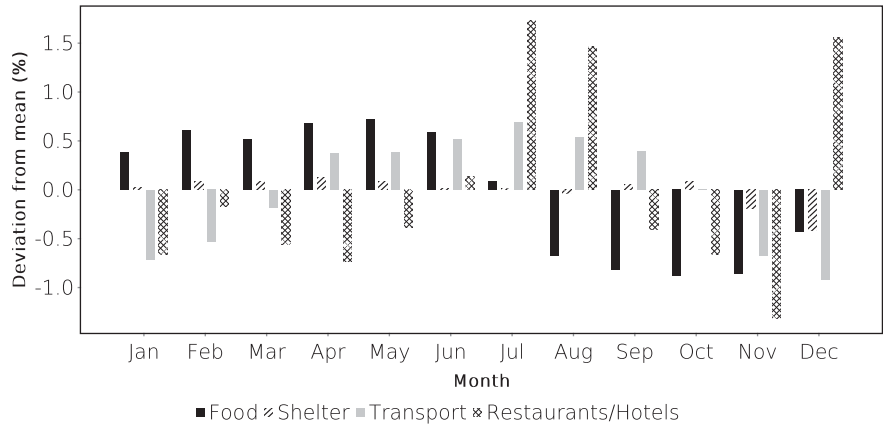


FIGURE 23.5 Seasonality factors (percentage deviations from trend developments) for the 4 main components of German HICP, ordered by their weight in the overall HICP index.

the seasonality factors of the 4 main components in German HICP, extracted in the same way as in Figure 23.4 from HP-filtered data. Notably, the German HICP component related to restaurant food and hotel stays shows clear peaks in the main holiday periods of summer and Christmas. Although the weight of this component in the overall HICP is less than 10%, the high volatility of this sub-index means that it contributes significantly to the overall seasonal pattern in the overall index. This observation explains the importance of using somewhat more sophisticated time-series filters than the HP filter used here. Movable feasts can affect highly volatile components of inflation at different times of the year and simple filters will treat these predictable effects as noise instead.

At the same time, highly volatile inflation index components raise an important concern about price indices. In some case, prices follow demand and the spikes in hotel prices during holiday seasons are probably due to such demand peaks. Other price fluctuations are driven by supply, however. For instance, higher demand for light oil distillates such as gasoline during the summer driving season tends to create higher supply in heavier distillates such as heating oil and diesel because a given barrel of crude oil contains a fixed mix of both (see [54] for a good introduction to this topic). Many consumers are aware of attendant seasonal price fluctuations and are able to adjust their consumption schedules to these patterns. For instance, European households that use heating oil tend to buy this fuel in summer when it is cheap because they have sufficient storage capacity to carry the oil into the winter season. The inflation index, however, assigns a fixed weight to the heating component of the sub-index related to shelter. The effect of this fixed weight is that the absolute level of heating costs is over-estimated in the price index. However, because households are unable to carry substantial amounts of fuel over several years, the rate of change in the price index is unlikely to be significantly affected as long as the seasonal price and consumption patterns remain unchanged⁷.

The value of an inflation-linked bond is given by its dirty price multiplied by the index ratio R_t . Given that seasonality of inflation leads to a seasonal pattern in R_t , and that short-term expected returns on bonds should follow short-term interest rates which do not have such a seasonal pattern, the clean prices of inflation-linked bonds must somehow absorb that seasonality. Before going into the theoretical details, one can look at empirical evidence. Figures 23.6 and 23.7 use a simple cyclical cubic smoothing spline (cf. e.g. [87]) fitted to the clean prices of every inflation-linked bond in their market. The prices of each individual bond have been normalised to 100 at the start of each year to reduce the impact of coupons on the absolute price range.

The fit does not address any non-seasonal patterns such as changes in long-term inflation expectations. Such patterns can be assumed to approximately average out over the observation periods used here (around 10 years for the US and 7 years for the German data).

The charts suggest that by and large the market for inflation-linked bond rationally reflects seasonality in the underlying index through adjustments in the (not inflation-adjusted) clean prices. However, while the amplitude of seasonal price fluctuations in the US more or less matches the swing in the seasonality, the price swing is much larger in German bonds.

Because the clean prices affect real yield calculations (addressed below), and real yields drive breakeven inflation, breakeven inflation also has a seasonal pattern. This

⁷As an aside, the ability to engage in this form of strategic consumption timing is tied to sufficient idle assets. Only consumers who are not constrained by liquidity can choose to purchase items when they are cheap. Others will purchase these goods as and when they are required, subject to necessary funds. This suggests that the absolute level of prices paid for the same goods can be lower for consumers with sufficient wealth than for less wealthy members of society, at least as long as price patterns are reasonably predictable. Whether this is a social problem is debatable given that the contents of consumption baskets are also determined by wealth and the total consumption tends to increase with income.

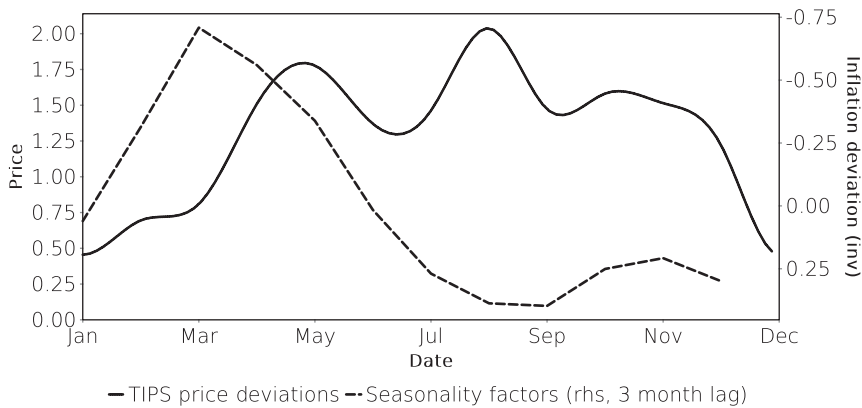


FIGURE 23.6 Seasonal patterns in the clean prices of US TIPS and lagged seasonality of CPURNNSA (from Figure 23.4). The cyclical cubic smoothing spline is fit to 81,500 observations taken from treasurydirect.gov. Source: Data obtained from treasurydirect. Gov.

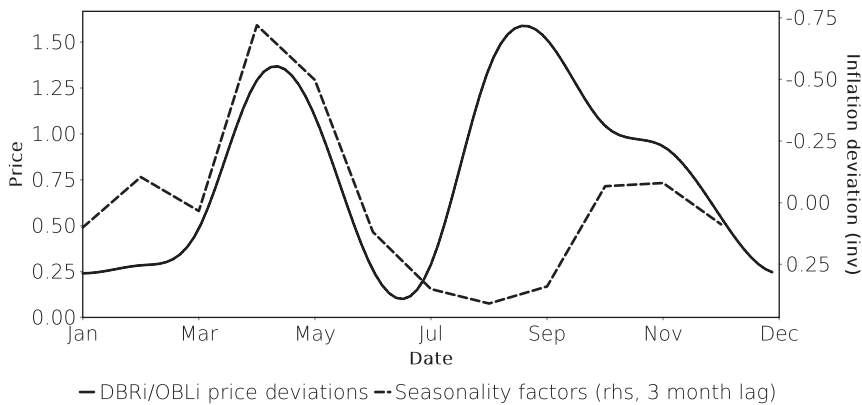


FIGURE 23.7 Seasonal patterns in the clean prices of German DBRi and OBLi prices with the lagged seasonality of HICP (cf. Figure 23.4). The cyclical cubic smoothing spline is fit to 8,500 observations taken from the Bundesbank. Source: Data obtained from the Bundesbank.

realisation is important for the interpretation of breakeven inflation as a measure of long-term inflation expectations.

23.4 PRICE FORMATION IN INFLATION-LINKED MARKETS

The relatively stable seasonality patterns in inflation indices and their influence on the pricing patterns of inflation-linked fixed income products can provide some clues on the price discovery process.

Figure 23.8 shows the seasonality in zero-coupon inflation swap rates, extracted in the same way as elsewhere in this chapter through the comparison of a

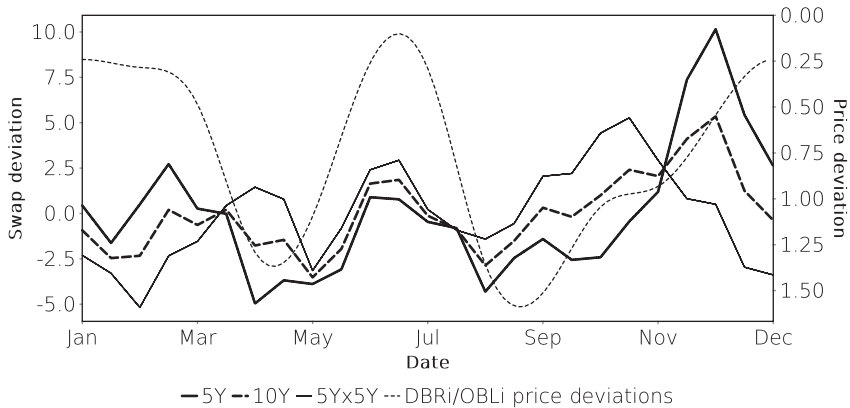


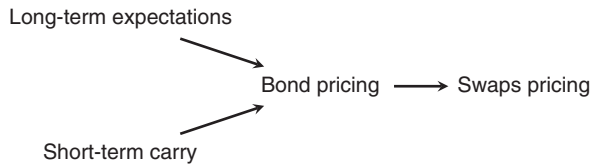
FIGURE 23.8 Seasonal patterns in the clean prices of German DBRi and OBLi prices (inverted) with the seasonality of HICP swaps. Sources: Data from ECS SDW, Bundesbank.

Hodrick-Prescott-filtered time series with the original data. The inflation swaps cover an integer number of years and should therefore not incorporate any seasonality. Furthermore, the chart shows not only the spot-starting swap rates but also the 5Yx5Y forward inflation swap rate⁸. This forward rate should not only not exhibit seasonality, but should also be covering inflation expectations beyond the forecast horizon of realistic economic observers. Despite this, the forward rate has a seasonal pattern is driven more or less by the two spot rates. To some extent, this pattern also seems to correspond to the one observed for index-linked bond prices.

A possible explanation for this somewhat non-intuitive outcome is that the breakeven inflation rates seen in the market are driven at least partially by short-term considerations, rather than solely by rational expectations of future realised inflation. In particular, the inflation swap market differs from the inflation-linked bond market in that transactions are generally symmetrical in that for each inflation payer in the swap market there is an equal receiver. The bond market, meanwhile, has a relatively constant outstanding volume while the market appetite for such bonds waxes and wanes over time. For speculative holders of inflation-linked bonds, short-term income is an important consideration because limits on P&L drawdowns usually prevent traders from holding losing positions for long periods. A trader with a firm view on inflation five years from now will find it difficult to express such a view with a 5Yx5Y inflation swap because a 10-year investment is beyond the usual career planning horizon in most financial firms. At the same time, the inflation-linked bond market provides a natural source for paying interest in the inflation swap market in that a trader can buy an inflation-linked bond, pay fixed in the nominal swap market to hedge out the real coupons, and pay inflation in the zero-coupon inflation-swap market to offset the index link of the bond.

⁸In line with market convention, this rate is calculated from the 5Y and 10Y zero coupon rates r_5 and r_{10} with the approximation $r_{5x5} = ((1 + r_{10})^{10} / (1 + r_5)^5)^{0.2} - 1$.

This market structure then implies that short-term developments in the carry on inflation-linked bonds drive their pricing to some extent, and so interfere with the rational expression of long-term inflation expectations. Then main source of noise in these short-term carry developments are related to changes in the oil price because energy prices are at the same time volatile and have a high weight in the inflation indices. Because traders generate inflation swap paying interest from the inflation-linked bond markets, seasonal and short-term fluctuations affect inflation swaps although the swap rates, particularly the longer-dated forwards, have no fundamental reason to reflect these fluctuations.



As a result, inflation seasonality and changes in energy prices affect ostensibly long-term measures of inflation expectations. The extent to which this happens depends on a number of other market factors, such as the general availability of arbitrage capital, that are difficult to quantify.

23.5 RETURN MEASURES OF INFLATION-LINKED BONDS

Even allowing for the fiction that the concept of yield for nominal bonds reflects the reasonable assumption of a constant rate at which cashflows can be reinvested, inflation-linked bonds present a problem. Inflation is clearly not fully predictable (otherwise inflation-linked bonds need not exist), so the inflation uplift experienced by reinvested real coupons is uncertain.

The first solution is to simply ignore the idea of inflation altogether and treat Equation (16.10) as referring to the real coupons and the real (not inflation-adjusted) price of the respective inflation-linked bond. This gives rise to the idea of the real yield ρ of inflation-linked bonds.

At this point, it is useful to refer back to the assertion that TIPS-style inflation-linked bonds can be treated, mathematically at least, like foreign currency bonds. In this setting, the real yield is the equivalent to the yield of the bond calculated in its domestic market. In other words, if a USD-based investor were to purchase a JGB, the yield of the JGB calculated the normal way in Japanese Yen would play the role of the real yield of a TIPS, while the nominal yield of the equivalent US Treasury bond would be the best estimate of the USD return of that JGB. Both return measures have no direct bearing of the return in nominal terms for the investor because FX or inflation risk, respectively, would introduce additional return components.

If an investor is instead searching for a nominal return measure for inflation-linked bonds, one solution is to forecast inflation up to the maturity of the bond, calculate the expected nominal cashflows based on the resulting values for I_t on the relevant cashflow

dates and calculate the yield using Equation (16.10). Naturally, this yield forecast will depend on each individual investor's inflation forecasts. In a world of efficient markets and risk-neutral pricing, however, this effort will be in vain because the best possible estimate for the nominal return r corresponding to the investment horizon of the respective inflation-linked bond is already known, namely, it is the yield of the nominal bonds of the relevant horizon. One therefore postulates the existence of a nominal yield of inflation-linked bonds and assumes it to be equal to the (possibly interpolated) yield of the comparable nominal bonds.

Of course, some of the assumptions stressed here are totally unrealistic and so nominal yields of inflation-linked bonds are to be interpreted with caution. Turning back again to the foreign currency analogy and the USD/JPY example, it is indeed possible to make an efficient market argument that the expected unhedged return on a 10Y JGB investment for USD-based investors is equal to the 10Y Treasury yield. However, given the heroic assumptions underlying this argument, few would elevate this return measure to the status of a yield which, though flawed (cf. Section 16.7), is at least based on a firm mathematical formula.

23.6 BREAKEVEN INFLATION

Arguably, the main use of the nominal yield of an inflation-linked bond is as an input for the calculation of breakeven inflation. Simply speaking, breakeven inflation ι is the rate of inflation that will make the internal rate of return of investing in an inflation-linked bond identical to that of a nominal bond of identical maturity.

There are 3 usual methods to express breakeven inflation. They are given below together with mnemonics referring to well-known relationships in other areas where they exist.

Fisher hypothesis The simplest, and most common, calculation of breakeven inflation $\iota = r - \rho$ corresponds to the idea that nominal rates are the sum of growth and inflation expectations. The main drawback is that this equation neglects any compounding effects.

FX points Using the real and nominal interest rates as interest rates in two different currencies, breakeven inflation can be interpreted as the annual change in the exchange rate that will make the returns in either currency identical. Breakeven inflation is therefore given by the well-known formula for FX forward points:

$$\iota = \frac{1 + r}{1 + \rho} - 1 \quad (23.9)$$

used here with the tacit assumption of a 1-year horizon.

Exact The exact way to calculate breakeven inflation assuming a constant rate ι is to use a given ι , calculate the resulting nominal cashflows from the inflation-linked bond and then calculate the internal rate of return of these cashflows using the equivalent of Equation (16.10). Then, ι is adjusted until the internal rate of return so obtained matched the nominal yield of the index-linked bond. This corresponds to

a two-stage iteration process (the calculation of y in Equation (16.10) is an iteration process, and this iteration has to be repeated for each trial value of t). Unless this calculation is done using a proper seasonality adjustment, the added precision of this calculation is somewhat illusory.

Markets generally use the first definition but data providers also offer the third. A consequence of the different risk characteristics of inflation-linked and nominal bonds is that breakeven inflation is directional with the level of nominal yields.

Breakeven inflation as calculated here suffers from seasonality effects when used as an indicator of inflation expectations. Because bond returns are measured between settlement date and maturity, settlement on a date when the reference index is seasonally low would push breakeven inflation upwards, while a maturity date on a seasonally low breakeven index would push it downwards. Simply speaking, breakeven inflation should measure the average annual rate of change in the underlying inflation index between settlement and maturity, subject to the interpolation lag (cf. Figure 23.1). If the intervening period between these 2 dates is not an integer number of years, seasonality can distort this interpretation.

Figures 23.9 and 23.10 show this effect in a very simple way. They are based on a constant HICP ex-Tobacco inflation of 2% with the historical seasonality pattern added to the observed index levels. In the first example, two different end dates differing by three months are chosen and the average inflation rate between a common settlement date and these two maturity dates are shown⁹.

The average inflation to the maturity date falling on a seasonally high index reading appears higher although the underlying inflation path is the same as for both bonds.

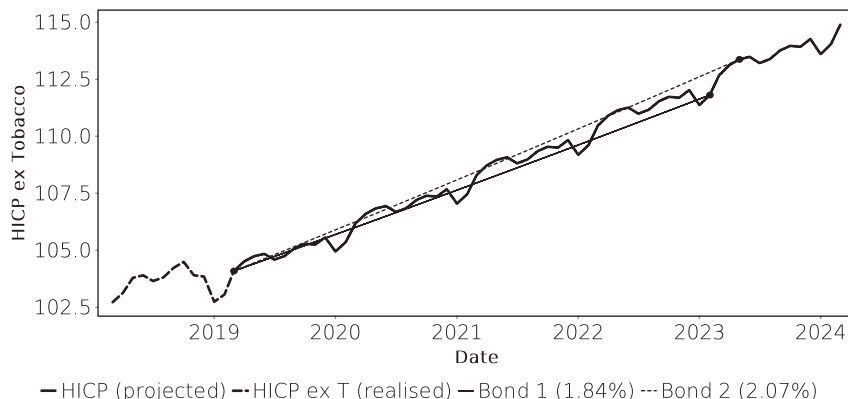


FIGURE 23.9 Effect of seasonality on breakeven inflation. Average annual inflation is 2% in this example but the seasonality of HICP ex-Tobacco is added to the projected inflation path. When the maturity date falls on a seasonally low index level (Bond 1), breakeven inflation appears to be lower.

⁹This chart, and the next, ignores the inflation lag for simplicity.

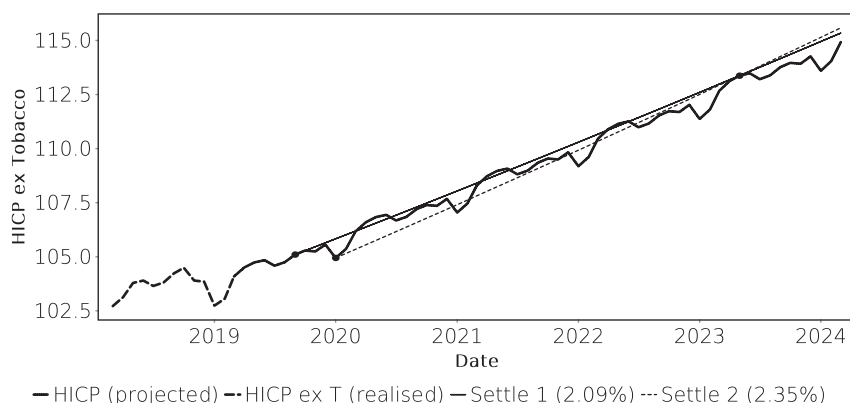


FIGURE 23.10 Effect of seasonality on breakeven inflation. Average annual inflation is 2% in this example but the seasonality of HICP ex-Tobacco is added to the projected inflation path. When settlement occurs on a seasonally high index level (Settle 1), the apparent breakeven inflation rate is lower.

This effect complicates the comparison of bonds linked to the same index but maturing in different months, such as different Japanese inflation-linked government bond series, or between French and German government bonds linked to HICP ex-Tobacco.

At the same time, even for an unchanged inflation path breakeven inflation can change as a result of changing settlement days. In Figure 23.10 the maturity date and underlying inflation is the same but the settlement date changes. The average inflation rate between settlement and maturity changes significantly even though the actual underlying annual inflation remains constant.

The effect of seasonality on breakeven inflation diminishes with increasing bond maturity because the same absolute change in the price level is amortised over a larger time horizon. Figures 23.9 and 23.10 were drawn for relatively short maturities to make the seasonality effect more visible. Also, the reason why Figures 23.6 and 23.7 use the price, not yield, changes of bond prices throughout the year is that the seasonality effect is an absolute price shift which has a smaller effect on the yields of longer-dated bonds.

An important question is whether breakeven inflation is in any way related to inflation expectations. In theory, the expected inflation rate over the lifetime of an inflation linked bond should be decisive in deciding whether to buy a linker or the equivalent nominal bond. In practice, this argument assumes a balanced market and inter-temporal substitution of future returns. An investor buying an inflation-linked bond at a time of low inflation because she or he expects inflation to rise will initially underperform an investor in the nominal bond. Until the time this asset allocation decision is vindicated by either a change in market prices (higher breakeven inflation) or realised returns (higher index ratio), this underperformance needs to be justified. Doing so is complicated by the ready convertibility of nominal into inflation-linked bonds which argues against early hedging of rising inflation¹⁰. In addition, the amount

¹⁰The vernacular summary of this argument is ‘Nobody buys an umbrella until it rains.’

of inflation-linked bonds circulating in the market is independent of the demand for them given that developed-market inflation-linked debt has long maturities¹¹. At times, there may be a surfeit of such bond circulating, which will depress their prices and thereby breakeven inflation. At other times, investors may be scrambling to find real assets for fear of imminent inflation, and breakeven inflation may be elevated relative to expectations because short-term fears override long-term expectations.

Put differently, using breakeven inflation of bonds as a measure of future expected inflation is equivalent to relying on the rationality of others, and their willingness to risk capital in the expression of their views. The deviations between inflation expectations, and market-based inflation rates arising from the cost of expressing expectations are known as the inflation risk premium.

23.7 CARRY ON INFLATION-INDEXED BONDS

Carry on inflation-linked bonds at first sight is no different from the carry on nominal bonds. For an initial cash outlay of P_0 at t_0 , the arbitrage-free forward nominal cash price of the bond at t_f is $P_f = P_0/Df(t_0, t_f)$. The complication arising is that inflation-linked bonds are usually not quoted in nominal, but real terms. In other words, the invoice price P_0 is given by $P_0 = R_0\hat{P}_0$ where $\hat{P}_0$ is the quoted real price (including accrued interest). Correspondingly, $P_f = R_f\hat{P}_f$ and so:

$$\hat{P}_f = \frac{R_0}{R_f} \frac{\hat{P}_0}{Df(t_0, t_f)} \quad (23.10)$$

This expression allows some insight if one writes $R_f = (1 + DCF(t_0, t_f)i(t_0, t_f))R_0$ and also expands $Df(t_0, t_f) = 1/(1 + r(t_0, t_f)DCF(t_i, t_f))$. The rate $i(t_0, t_f)$ here is used as an interest rate but is in fact the (lagged) inflation rate between t_0 and t_f . This gives:

$$\hat{P}_f = \frac{1}{1 + DCF(t_0, t_f)i(t_0, t_f)} \hat{P}_0(1 + r(t_0, t_f)DCF(t_i, t_f)) \quad (23.11)$$

and for small values of $DCF(t_0, t_f)$, $i(t_0, t_f)$, and $r(t_0, t_f)$ this can be simplified to:

$$\hat{P}_f = \hat{P}_0(1 + (r(t_0, t_f) - i(t_0, t_f))DCF(t_i, t_f)) \quad (23.12)$$

This is formally very close to Equation (16.20) but instead of the repo rate $r(t_0, t_f)$ one has the strange expression $r(t_0, t_f) - i(t_0, t_f)$, i.e., the difference between the repo rate r and the inflation rate i . This rate is called the real short rate. Unfortunately, some economists have the tendency to sportingly call the difference between practically any inflation rate and any nominal interest rate real rate, so this term may be confusing.

¹¹Because near-term inflation can be predicted with reasonable accuracy, there is no need to buy inflation-linked bonds to hedge it, and hence no natural demand for short-dated inflation-linked bonds.

However, the meaning of $r(t_0, t_f) - i(t_0, t_f)$ should be obvious. If there was a market for short-term funds in which payments are adjusted according to inflation in the same way as for inflation-index bonds, one could use Equation (16.20) directly and would not have to worry about the inflation-linked nature of the bonds. As it is, one needs to use nominal repo interest rates and adjust for inflation explicitly.

So far, these formulas have ignored a little technical problem, namely that there is no general way to know either R_f or $i(t_0, t_f)$. Note that these are hard quantities in the sense that they are defined by the bond issuer and the respective statistics office. They are not things that can be decided by negotiation market counterparties, as for instance $r(t_0, t_f)$. There is no solution to this problem, but what is generally done is to use forecasts of I_f or $i(t_0, t_f)$ to estimate $\hat{P}_f$. This means, of course, that market participants have no problems agreeing on P_f for an inflation-linked bond (because they have to agree only on the repo rate $r(t_0, t_f)$ which is likely to be easily observed), but in order to agree on $\hat{P}_f$ they would have to agree in their inflation outlook as well. In many developed markets, there are forward contracts for I_f which can be used to hedge this risk, usually for up to about one year.

23.8 COMPREHENSIVE INFLATION MODELLING

The starting point of any model for the valuation of inflation-linked instruments is the nominal discount curve. Nominal rates products are more liquid than index-linked ones, and monetary policy, which ultimately governs the funding costs in an economy, is implemented in nominal terms.

Given the strong influence of seasonality on pricing, simple measures such as yields or breakeven-inflation do not suffice to analyse the relative valuation of index-linked bonds and inflation swaps. The only way to encapsulate all available information is to construct and calibrate a comprehensive model of the relevant inflation index in the future, and then use the nominal rates curve to discount these projected nominal cash flows. A very simple model for the inflation index I_f at a future date f has the form:

$$I_f = I_t \frac{s_f}{s_t} e^{\iota(f-t)} \quad (23.13)$$

where s_f and s_t are the seasonality factors applicable to settlement at spot (t) and the forward date f . The rate ι is essentially a zero-coupon breakeven inflation rate and the exponential term in which it is incorporated reflects the geometric nature of a fixed inflation rate in terms of the development of the underlying index. One would normally specify this model in terms of the monthly index readings, and then apply the lag and interpolation rules pertaining to the bonds in question.

If sufficient data is available, it is possible to make ι time-dependent (i.e., introduce a function $\iota(t)$) with a reasonably simple functional form rather than using a constant. This avoids a jump in the overnight inflation rate from its current value to the new constant ι . For instance, one could model a gradual convergence of the inflation rate to

a long-run value with the two-parameter function:

$$I_f = I_t \frac{s_f}{s_t} \exp \left((f - t)(\iota_0 + \iota_t) \frac{(f - t)/\tau}{1 + (f - t)/\tau} \right) \quad (23.14)$$

The long-run inflation rate would be $\iota_0 + \iota_t$ in this model, where ι_0 is the most current reading of the annual inflation rate. The time scale parameter τ defines a convergence speed from the current inflation but although this parameter is useful for curve fitting, it has little economic significance. In particular, when $|\iota_t| \ll 1$, the actual value of τ has no impact. Examples of the evolution of the assumed annual inflation rate and the projected index are shown in Figures 23.11 and 23.12.

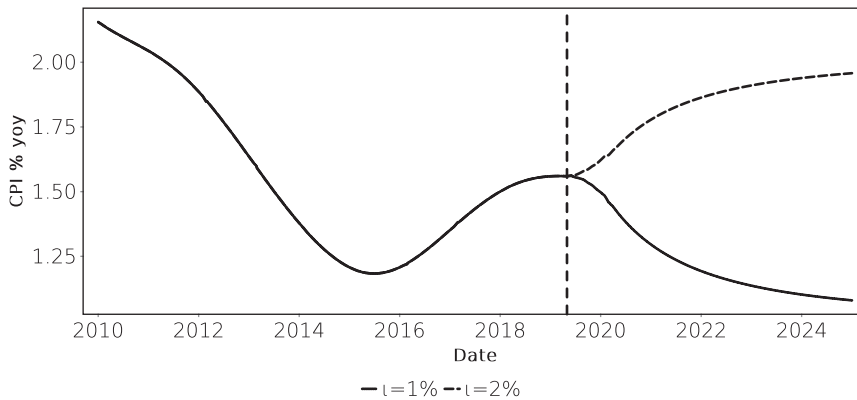


FIGURE 23.11 Examples of inflation rate dynamics of the model Equation (23.14). The two curves separate at the last known reading of the actual inflation index.

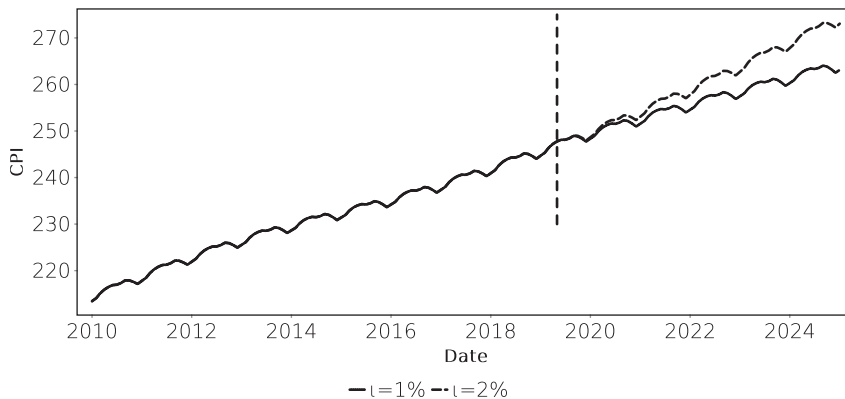


FIGURE 23.12 Examples of inflation projections for the purpose of modelling inflation-linked bond prices using two different ι values and $\tau = 3$ in Equation (23.14). The 2 curves separate at the last known reading of the actual inflation index.

Time-dependent ι assumptions are particularly important when there are future events that lead to jumps in the inflation index, for instance as a result of announced or expected tax increases¹². In more developed markets, the exigencies of repo trading in inflation linked bonds have created a more or less liquid market in near-term index forwards extending about 1 year into the future. Such forwards permit the extension of the starting point of the extrapolation Equation (23.13) into the future.

Having modelled the nominal cash flows, one can then use the standard equation linking them to the nominal spot price of the corresponding bond:

$$P = \sum_i Df(t_i) C f_i \quad (23.15)$$

Using the definition in Equation (23.13), for example, this can be re-written (with an appropriate change of variables) as:

$$P = \sum_i Df(t_i) \hat{C} f_i \frac{I}{I_0} \frac{S_{t_i}}{S_0} e^{\iota t_i} \quad (23.16)$$

This only holds if the bond in question has no indexation floor at I_0 for either coupons or principal, or if that floor is well out of the money. If the floor exists and is close to the money, then the cash flow at time t_i can be higher than the linear function shown here.

Equation (23.16) has a very useful form because the ι term is unrelated to the structure of a particular bond. By introducing a real discount factor:

$$Rf(t) = Df(t) \frac{S_t}{S_0} e^{\iota t} \quad (23.17)$$

Equation (23.16) becomes:

$$P = \frac{I}{I_0} \sum_i Rf(t_i) \hat{C} f_i \quad (23.18)$$

The analytical significance of this re-casting is that $Rf(t)$ can be represented as a composite curve model along the lines of Section 19.4.5 (page 202). A modelling hierarchy for an inflation-linked bond curve would therefore start with a spline for the nominal curve, and then stack an inflation model along the lines of Equations (23.13) or (23.14) with appropriate seasonality corrections on top of that nominal model to arrive at an appropriate real discount curve. A standard curve fitting algorithm can then use the ι parameter, or the two parameters (ι, τ) to match observed market prices. This has been done to produce the forward rate curve shown in Figure 23.13. Note that the forward rates show a strong seasonal pattern which is essentially the annualised slope of the seasonality of the underlying consumer price index. Such behaviour of the real rate contradicts the economic intuition of interest rates as consisting of a real return expectation plus some inflation compensation. However, seasonality in the index ratio calculation exists but nominal rates do not show a seasonal pattern. The real rate curve is therefore simply the only place where seasonality can be reflected in a yield curve model.

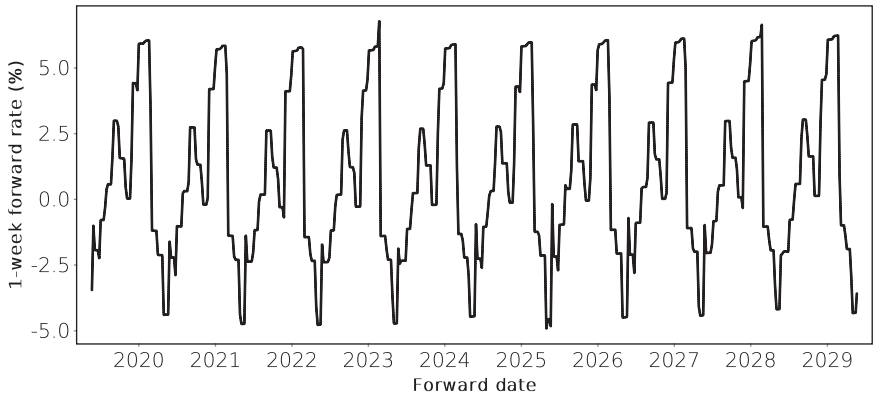


FIGURE 23.13 Real 1-week forward rates in a seasonally adjusted model using a standard Nelson-Siegel spline for nominal US Treasuries and the inflation model Equation (23.14) on 20 February 2020. Source: Prices from TreasuryDirect.

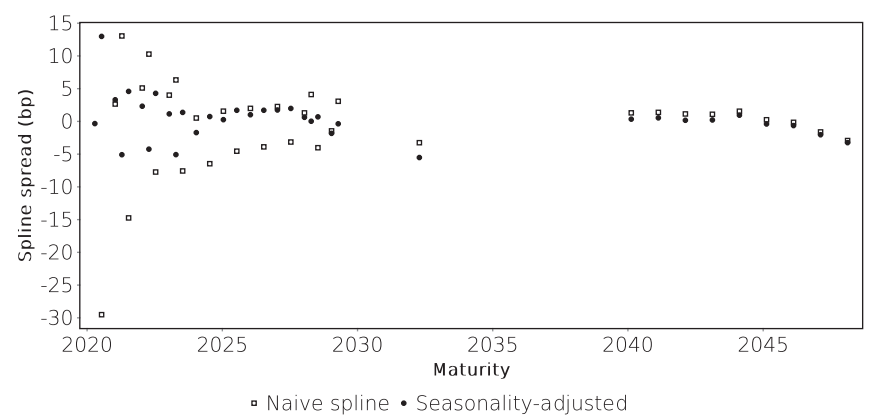


FIGURE 23.14 Spline spreads of US TIPS against a naive (not seasonally adjusted) and a comprehensive (seasonally adjusted) curve model. Source: Data from TreasuryDirect, inflation modelling using the model above.

With this apparatus of a real discount curve that reflects inflation expectations as well as seasonality, it is possible to construct an accurate rich-cheap model for individual inflation-linked securities. Figure 23.14 of US TIPS spline spreads shows that this approach is warranted. The July maturity TIPS appear substantially overvalued in a naive (not seasonally adjusted) model, but that is because they pay on the comparatively high April CPI reading. The April maturities, paying on the January CPI reading, look cheap in comparison. This effect was visualised conceptually in Figure 23.9 and

¹²The consumption tax hikes in Japan in 2014 and 2019 were examples where a functioning inflation-linked bond market had to properly reflect one-off jumps in the price level.

is here demonstrated with actual bonds. The solid markers are the spline spreads calculated with the seasonally adjusted real curve of Figure 23.13 and the substantially smaller range of spreads demonstrates the impact of seasonality on market valuations. The market does appear to retain some deviations from fair value even in this model, but that reflects the parsimony of the approach taken in Equation (23.18). New TIPS have a higher value of their deflation floor but the model does not reflect this value. It may also be that the market consensus seasonality model is different from the one used here (a Hodrick–Prescott filter using data since 2000).

23.9 INFLATION MODELS AND EXPECTATIONS

The discussion of the preceding section centred on the appropriate structure of an inflation model for the relative valuation of individual inflation-linked securities. An ancillary result of this modelling effort is a long-run inflation estimate and it can be interesting to treat it as an economic forecast. It is not immediately obvious that this model parameter should be a useful economic estimate. First, market prices contain risk premia in addition to pure expectations. In the absence of an explicit model for the risk premia one needs to consider this model estimate as biased. Second, if the number of observable inflation-linked bonds is small, the estimate of the long-run inflation rate can be subject to a large measurement error. In essence, each bond provides one sample for future inflation developments and only a large set of samples can make the aggregate estimate reliable. Third, unlike the inflation swap curve, the two models presented in Equations (23.13) and (23.14) do not have any pretence of having a meaningful term structure. Instead, the term structure of real yields in these models is inherited largely from the respective nominal yield curve¹³.

With these caveats in mind, Figure 23.15 shows the history of the long-run inflation estimate obtained from US TIPS prices. Each daily data point is the result of fitting splines to the nominal US Treasury curve and then calibrating a TIPS model using the definition Equation (23.14). The result suggests that inflation priced into the TIPS market is reasonably well anchored to the Fed's inflation target and fluctuates only mildly around 2%. This, from a central bank's point of view, benign picture only changed with the unprecedented decline in energy prices in March 2020.

For European inflation, the German linker market can be used to obtain a similar estimate but the smaller number of bonds makes this estimate more subject to volatility¹⁴. Figure 23.16 shows these estimates together with the HICP inflation swap 5Yx5Y forward rate which should in theory be a comparable estimate of long-run steady-state inflation¹⁵. Investors should be indifferent to holding German nominal

¹³It is of course debatable whether the term structure of inflation swaps provides any extra information about inflation developments or is also simply based on the relative levels of the real and nominal curves.

¹⁴The French OAT€i market would provide better estimates.

¹⁵There should be a convexity adjustment applied to the inflation swap forward. See footnote on page 239.

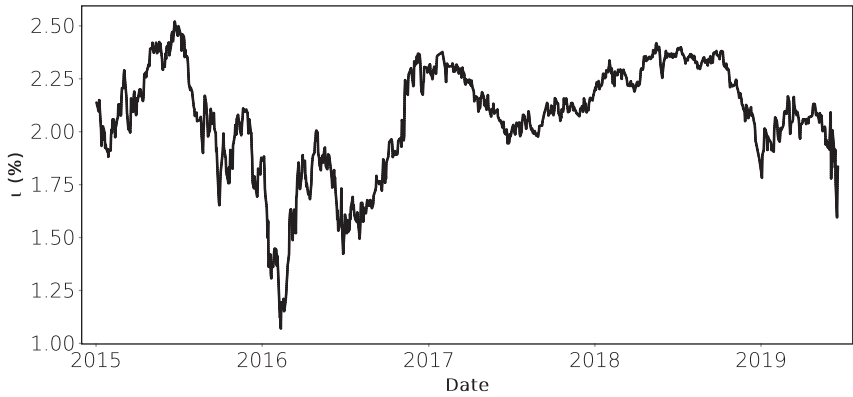


FIGURE 23.15 History of the long-run inflation obtained from fitting a TIPS spline. Source: Data from TreasuryDirect, inflation modelling using the model above.

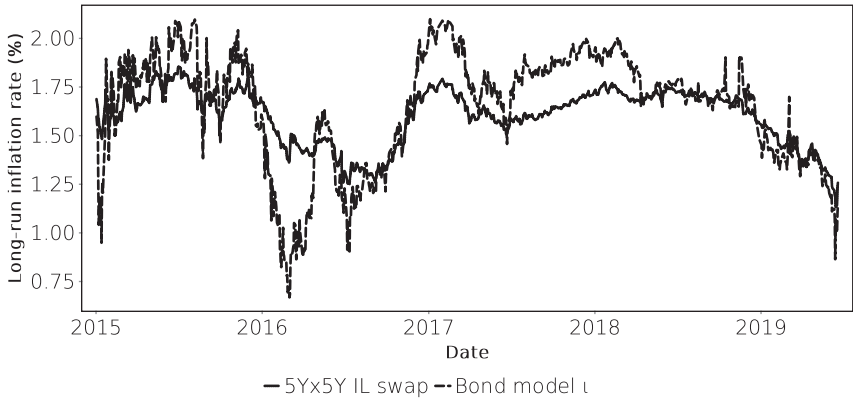


FIGURE 23.16 Model-based long-run HICP ex-tobacco estimate from German OBLi/DBRi prices and 5Yx5Y forward inflation swaps. Source: Prices from ECB SDW and Bundesbank.

bonds, and swapping the coupons into inflation, or holding inflation-linked bonds directly. The two measures indeed show a remarkable degree of correlation but also phases of divergence. If one subscribes to the pricing hierarchy shown on page 240, then the degree to which the bond-based inflation measure diverges from the swap measure would be an indication of the width of the arbitrage corridor around this equivalence. The average absolute deviation between the two series in the figure is about 15 bps which implies an arbitrage corridor of around 30 bps. This may not be an unrealistic estimate for the implied balance sheet cost of carrying an inflation-linked bond asset swap position that needs to be held for a long time to accrue an arbitrage profit. It also implies that either estimate of long-run inflation faces a comparable degree of uncertainty.

This should not be seen as a significant problem because the measurement of the price level itself is also subject to measurement errors. The German statistics office aims for a 0.1% percentage point accuracy [31] in its measurement of the German HICP. In other words, assuming that there is a consensus path of HICP inflation, then it appears that financial markets indicators of this path may be deviating by around 15 bps from this path due to the cost of arbitraging the relevant instruments. At the same time, the published levels of inflation may also be deviating from its true level by up to 10 bps.

